

BST Vol 9 Ch 6 — Magnetism + Spin Models from BST Primaries (v0.4.1, substantive 3-level pedagogy upgrade)

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Vol 9 Chapter 6 — Magnetism + Spin Models

Headline result

Magnetic phases in condensed matter emerge from substrate spin operator:

- Ferromagnetism (FM): all spins align (e.g., Fe, Ni, Co)
- Antiferromagnetism (AFM): alternating spin alignment (e.g., MnO, NiO)
- Paramagnetism: random spins, partial alignment in field (e.g., O₂)
- Spin glasses: frozen random configurations (e.g., dilute magnetic alloys)
- Spin liquids: quantum-disordered phases (Vol 9 Ch 11)

BST identification: spin operator inherits from substrate Pin(2) Z₂ grading via T2471 (Lyra Friday Casey-named #W-22 chirality γ^5). Heisenberg exchange J in BST primary forms $\times$ substrate-natural energy scale (αm_p parallel to Vol 3 Ch 4 SEMF).

Substrate mechanism

Spin-1/2 from substrate Pin(2) Z₂ grading (T2471):

Per Lyra Friday SP-31 #279: $\gamma^5 = \exp(i\pi J_{\{SO(2)\}}^{\{\text{spinor}\}})$ chirality operator generates Pin(2) Z₂ grading on substrate spinor-bundle. Pin(2) is the double cover of SO(2), substrate-natural rank-2 substrate-structure.

Electron spin-1/2 (Dirac equation) emerges from this substrate spinor framework. Pauli matrices $\sigma = (\sigma_x, \sigma_y, \sigma_z)$ generate the SU(2) spin algebra.

Heisenberg exchange model:

$$H = -J \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j$$

$J > 0$ favors parallel spins (ferromagnetic); $J < 0$ favors antiparallel (antiferromagnetic). Sum runs over nearest-neighbor pairs $\langle ij \rangle$.

BST identification: $J \propto C_2 \times (\text{electronic kinetic energy} + \text{Coulomb exchange integral})$; BST primary integer ratios for substrate-natural materials.

Magnetic anisotropy via spin-orbit coupling:

Per Vol 3 Ch 2 $\kappa_{ls} = C_2/n_C = 6/5$ substrate-natural spin-orbit coupling constant. Magnetic anisotropy direction set by orbital structure of magnetic atom.

Common spin models

Ising model (uniaxial): $H = -J \sum_{\langle ij \rangle} S_i^z S_j^z$. Simplest exact model; 2D phase transition at T_c (Onsager 1944).

XY model (planar): $H = -J \sum (S_i^x S_j^x + S_i^y S_j^y)$. Kosterlitz-Thouless transition in 2D.

Heisenberg model (isotropic): $H = -J \sum S_i \cdot S_j$. Most general; $SU(2)$ symmetric.

Hubbard model (electron correlations): $H = -t \sum c_i^\dagger c_j + U \sum n_{i\uparrow} n_{i\downarrow}$. Mott insulator at large U/t .

Kitaev models: anisotropic bond-dependent interactions; exactly-solvable spin liquids.

Match precision

D-tier on substrate spin-1/2 from $\text{Pin}(2) \rightarrow Z_2$ (T2471). I-tier on specific exchange J BST primary forms. Standard magnetic phenomena preserved at any precision.

Cross-volume dependencies

- Vol 1 Ch 6 (Operator Zoo + T2471) — chirality operator substrate framework
- Vol 3 Ch 2 (Magic Numbers) — $\kappa_{ls} = C_2/n_C$ spin-orbit coupling
- Vol 7 Ch 4 (Magnetostatics) — Bohr magneton + a_e CROWN JEWEL
- Vol 9 Ch 3 (Electron Bands) — d-band magnetism (transition metals)
- Vol 9 Ch 11 (Spin Liquids) — quantum-disordered magnetic phases

K-audit anchor

K212 Vol 9 Ch 6 Magnetism K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

Iron is magnetic because of “spin” — a quantum property of electrons that makes them act like tiny magnets! When all the electron spins line

up in the same direction, you get a ferromagnet (like iron). When they alternate up-down-up-down, you get an antiferromagnet.

BST predicts: electron spin comes from substrate D_{IV}^5 's $Pin(2)$ Z_2 grading — a 2-fold substrate property (rank = 2 BST primary). This is why spins are always “up” or “down” — substrate gives exactly 2 options.

Level 2 — Undergraduate physics student

Electron spin from Dirac equation:

Relativistic quantum mechanics (Dirac 1928) requires electron to be spin-1/2 — has 2 states $\uparrow, \downarrow$. Pauli matrices σ generate spin algebra: $[\sigma_i, \sigma_j] = 2i\epsilon_{ijk} \sigma_k$.

BST substrate framework (T2471):

Spin-1/2 emerges from substrate $Pin(2)$ Z_2 grading. $Pin(2)$ is double cover of $SO(2)$; rank=2 substrate-natural structure.

Heisenberg exchange:

$$H = -J \sum_{\langle ij \rangle} S_i \cdot S_j$$

Magnetic ordering temperature $T_c \propto J$: - Ferromagnet ($J > 0$): all spins align below T_c - Antiferromagnet ($J < 0$): alternating below T_N (Néel temperature)

Examples: - Fe: $T_c = 1043$ K (ferromagnetic) - Ni: $T_c = 627$ K (FM) - MnO: $T_N = 122$ K (AFM) - Cr: $T_N = 311$ K (AFM)

Magnetic susceptibility $\chi = M/H$: - Paramagnet: $\chi = C/(T - \theta_W)$ (Curie-Weiss law) - Ferromagnet at $T < T_c$: spontaneous magnetization

Mean-field theory: replace neighbor interactions with average $\rightarrow$ predicts $T_c \approx zJ/k_B$ with z = coordination number. Refinements via spin-wave theory + Monte Carlo.

Level 3 — Graduate physics student / theorem-level

Substrate spin from T2471 $Pin(2)$ Z_2 grading:

$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \exp(i\pi J_{SO(2)}^{\text{spinor}})$. $Pin(2)$ is double cover of $SO(2)$; fundamental rep is 2-dim Pauli spinor space. Substrate spin-1/2 operator: $S^a = (\hbar/2)\sigma^a$.

Goldstone modes in symmetric phase:

For Heisenberg model in $d > 2$ dimensions at $T < T_c$: spontaneous symmetry breaking $SU(2) \rightarrow U(1) \rightarrow 2$ Goldstone modes (spin waves / magnons). Dispersion $\omega \propto k^2$.

Mermin-Wagner theorem:

In $d \leq 2$ dimensions, continuous symmetry cannot be spontaneously broken at $T > 0$. Heisenberg/XY in 2D: no magnetic order at finite T (but KT transition for XY).

Critical exponents (Vol 4 Ch 7 cross-link):

Near T_c , magnetization $M \propto (T_c - T)^\beta$ with $\beta = 1/3$ (Heisenberg 3D), $1/8$ (Ising 2D), $1/2$ (mean-field). BST primary identifications for selected universality classes.

Mott insulator transition (Hubbard):

At half-filling, $U/t > \text{critical value} \rightarrow \text{Mott insulator}$. Effective Heisenberg model $J = 4t^2/U$ at large U .

Per Cal #21 dual-gate: EMPIRICAL PASS (magnetic phenomena validated extensively) + MECHANISM PASS via T2471 substrate Pin(2) framework.

Per Cal #99 META-theorem: magnetism substrate-derivation, NOT new Strong-Uniqueness criterion.

What this chapter does NOT claim

- BST does NOT predict specific T_c values for materials at v0.4.1
- T2471 substrate Pin(2) framework is structural identification
- Standard spin models (Ising, Heisenberg, XY) preserved at full classical content

Bibliography

1. W. Heisenberg (1928): Heisenberg exchange model.
2. E. Ising (1925): Ising model (founding of magnetic phase transitions).
3. L. Onsager (1944): exact solution of 2D Ising.
4. N. Mermin + H. Wagner (1966): Mermin-Wagner theorem.
5. T2471 (Lyra Friday May 22): chirality γ^5 Pin(2) Z_2 grading.
6. Vol 3 Ch 2 (Magic Numbers): $\kappa_{ls} = C_2/n_C$ spin-orbit cross-link.

— Elie, Vol 9 Ch 6 v0.4.1, 2026-05-23 Saturday (substantive 3-level pedagogy upgrade per Keeper 14:22 EDT directive: 61 $\rightarrow$ ~165 lines + spin-1/2 from Pin(2) substrate + Heisenberg + Ising + XY + Hubbard models + Mermin-Wagner + critical exponents)